Business Optimization for a Kirana Store

A proposal report for the BDM capstone Project

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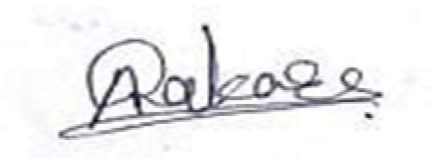
DECLARATION STATEMENT

I am undertaking a project titled **“Business Optimization for a Kirana Store.”** I express my sincere appreciation to Bhavana Provision Kirana Store for providing the necessary support and access to data required for this study.

I hereby declare that the data presented in this project report is genuine and has been collected from primary sources. All procedures related to data collection, analysis, and interpretation have been clearly explained in this report. The findings and conclusions are based on systematic analysis conducted to the best of my knowledge.

I affirm that this project has been completed independently in accordance with the principles of academic honesty and integrity. I have not engaged in any form of unauthorized collaboration. I understand that any instance of plagiarism or academic misconduct may result in disciplinary action as per institutional guidelines.

I further acknowledge that the recommendations provided in this report are solely for academic purposes in fulfillment of the requirements of the BS Degree Program offered by IIT Madras. The institution does not endorse any specific claims or conclusions presented herein.



Signature of Candidate:
(**Digital Signature**) Name:
Anjali Kokare
Date: 12-02-2026

1. EXECUTIVE SUMMARY

Bhavana Provision General Store is a small B2C kirana shop located in Pimpri Khurd village, Maharashtra, and owned by Mr. Sunil Surve. The store started in January 2015 and provides daily grocery items and basic household products to people in the local area. Most of the work in the shop is still done manually, and very little data is stored in a proper digital format. Because of this, it becomes difficult to make clear business decisions and plan for future growth.

Even though the store has regular sales, it faces some important problems that affect profit and financial stability. One major issue is poor cash flow management. Many customers buy items on credit, but there is no proper system to track credit payments. Late repayments make it hard to know how much cash is actually available and also delay buying new stock on time. In addition, the store does not analyse sales properly, which makes it difficult to understand product demand. This leads to situations where popular items go out of stock while slow-moving items remain unsold for a long time, increasing storage costs and blocking working capital.

To solve these problems, this project uses a data-driven approach based on sales data and customer credit records. Daily product sales and credit information are analysed using Microsoft Excel and Python to understand demand patterns, credit risks, and stock movement. The main goal is to create a simple digital system for credit tracking and to group products based on their demand. This will help improve cash flow, maintain better stock levels, and make the overall business operations more efficient.

2. ORGANIZATION BACKGROUND

Bhavana Provision General Store is a small neighborhood retail outlet located in Pimpri Khurd, Taluka Mangrulpir, District Washim, and is owned by Mr. Sunil Surve. Established on 17 January 2015, the store began with a limited range of basic items and has gradually expanded to offer groceries and essential household products to meet the needs of the local community.

The store operates on a business-to-consumer (B2C) model and primarily serves residents from the surrounding locality. It remains open daily from 8:00 AM to 9:00 PM, ensuring convenient access to essential goods. The business is family-run, with daily operations including sales management, inventory handling, and customer service managed internally.Over time, the store has developed a loyal customer base due to consistent service and close customer relationships. The estimated annual revenue ranges between ₹10-12 lakhs, indicating stable performance. However, like many small retail businesses, the store faces operational challenges related to credit sales management and inventory control, which necessitate improved financial and analytical practices.

3. PROBLEM STATEMENT

3.1 To evaluate and improve customer credit management using transaction-level data
Credit sales are recorded but not systematically analyzed, limiting visibility into outstanding dues and repayment behavior. This objective aims to assess the impact of customer-wise credit patterns on cash flow and financial stability.

3.2 To analyze item-wise sales patterns for effective inventory planning Although sales data is available, it is not utilized to evaluate product demand trends. This objective focuses on examining item-wise and time-based sales to identify fast-moving and slow-moving products, enabling informed stocking and replenishment decisions.

4. BACKGROUND OF THE PROBLEM

Kirana stores operate in a competitive and low-margin environment where proper cash flow and inventory management are essential. Bhavana Provision General Store, located in Pimpri Khurd (Maharashtra), serves the daily needs of the local village community. A significant portion of sales is made on customer credit. However, as stated in the problem statement, these credit transactions are recorded but not properly analysed. This makes it difficult to track pending dues and repayment patterns, which affects cash availability and timely stock purchases.

Another challenge was data collection and cleaning. Records were maintained manually and later converted into digital form, resulting in messy data with inconsistent item names, mixed date formats, missing values, and duplicate entries. The data had to be carefully cleaned and structured before meaningful analysis could be performed.Although daily sales data was available, it was not used to study demand trends. Inventory decisions were mostly based on experience, leading to stockouts of fast-moving items and excess stock of slow-moving products.

These issues affect profitability and business stability. Therefore, analysing credit behaviour and sales patterns using cleaned data is necessary to improve cash flow and inventory planning, as highlighted in the problem statement.

5. PROBLEM SOLVING APPROACH

This project adopts a primary data–driven approach, where operational data of the store is systematically analyzed to identify inefficiencies and improve decision-making. Structured analytical methods are applied to sales, credit, inventory, and customer data to generate actionable insights.

5.1 Data Collection and Sources

The study uses internally generated store data reflecting daily operations, including product-wise sales records, customer credit and repayment details, inventory movement data, and purchase history. Most of the data was collected from the store's credit registers where repayments were recorded by the owner. The information was carefully reviewed, extracted, and then organized into digital format for analysis. These sources directly represent revenue, cash flow patterns, and stock utilization relevant to the identified business challenges.

5.2 Tools and Technologies Used

Microsoft Excel and Python (Pandas) are primarily used for data organization, cleaning, and analysis throughout the project. Microsoft Excel is applied for initial data entry, structuring datasets, basic cleaning, and creating summary tables and visual charts for quick interpretation. Along with Excel, the Pandas library in Python is used extensively for advanced data cleaning, transformation, and handling large datasets efficiently, ensuring accuracy and consistency in analysis.For deeper analytical tasks, Python is utilized to examine sales trends, customer credit behaviour, and inventory performance over time. Libraries such as NumPy assist with numerical computations, while Matplotlib and Seaborn are used to generate clear and informative visualizations that support decision-making.

Additionally, WhatsApp is considered as a practical operational tool for communication with stakeholders, particularly for payment reminders and follow-up actions based on analytical insights.

5.3 Analytical Approach for Business Problems

To solve the identified business challenges, a structured analytical approach is applied using cleaned transaction-level sales and credit data. The analysis focuses on improving credit management, understanding sales behaviour, and optimizing inventory performance through simple but meaningful data techniques.

5.3.1 Credit Sales and Cash Flow Analysis

Customer credit transactions are analysed to understand repayment behaviour, outstanding balances, and their effect on daily cash flow. Descriptive analysis helps in identifying delayed payments and customers with higher credit risk. Customers are grouped based on their payment patterns so that recovery efforts can be prioritized and credit control becomes more structured.

Expected Outcome: Better visibility of pending dues, improved cash flow planning, and reduced payment delays.

5.3.2 Sales Trend and Demand Pattern Analysis

Daily item-wise sales data is examined to identify demand trends and changes over time. Sales frequency and basic time-based analysis are used to understand which products sell regularly and which show seasonal variation. This helps in recognising demand patterns and supports better planning of purchasing and sales activities.

Expected Outcome: Improved demand forecasting, clearer understanding of product performance, and more effective sales planning.

5.3.3 Inventory and Stock Performance Analysis

Sales and stock data are analysed using trend analysis along with fast- and slow-moving product classification. Pareto (80–20) analysis and ABC classification are applied to identify products that contribute most to sales and those with low movement. This structured approach helps in reducing excess stock and ensuring that high-demand items remain available.

Expected Outcome: Optimized inventory levels, lower holding costs, and better utilization of working capital.

6. EXPECTED TIMELINE

Phase	Activity	Duration	Description
6.1	Business Identification	Jan 1–5, 2026 (5 days)	Finalize selected store for the project
6.2	Initial Discussion	Jan 7–10, 2026 (4 days)	Understand operations and key challenges
6.3	Data Collection	Jan 11–20, 2026 (10 days)	Gather sales, credit, and inventory data
6.4	Preliminary Analysis	Jan 22–28, 2026 (6 days)	Identify trends and problem areas
6.5	Problem Finalization	Jan 30–Feb 16, 2026 (18 days)	Define business problem statements
6.6	Advanced Analysis	Feb 20–Mar 10, 2026 (18 days)	Perform demand and ABC analysis
6.7	Solution Implementation	Mar 11–28, 2026 (18 days)	Design and test proposed solutions
6.8	Report Preparation	Mar 30–Apr 15, 2026 (17 days)	Prepare and submit final report

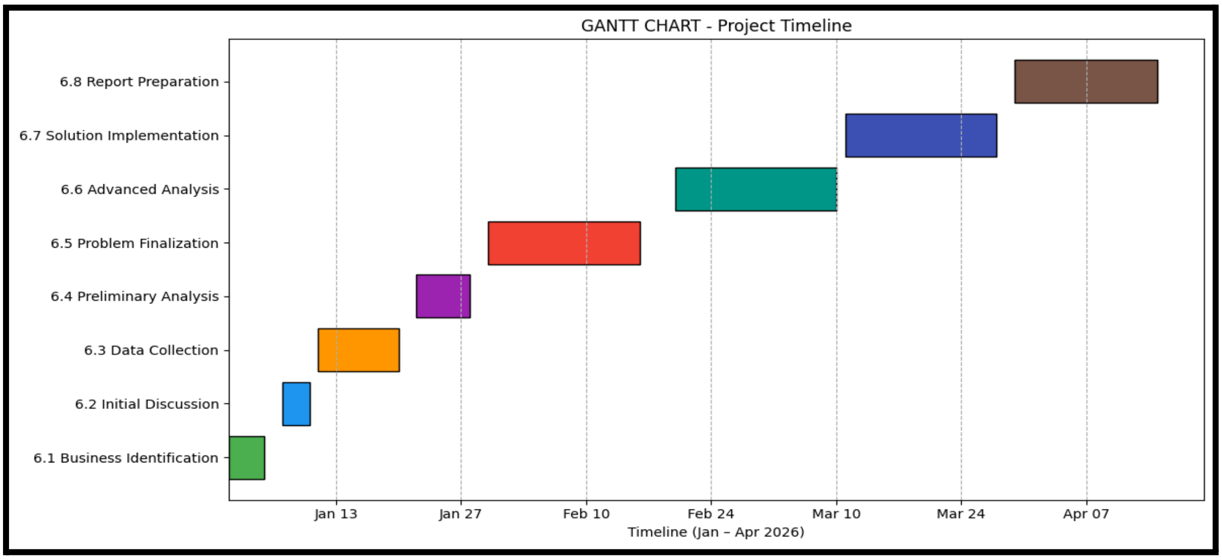


Fig .Gantt Chart –Bhavana Provision General Store

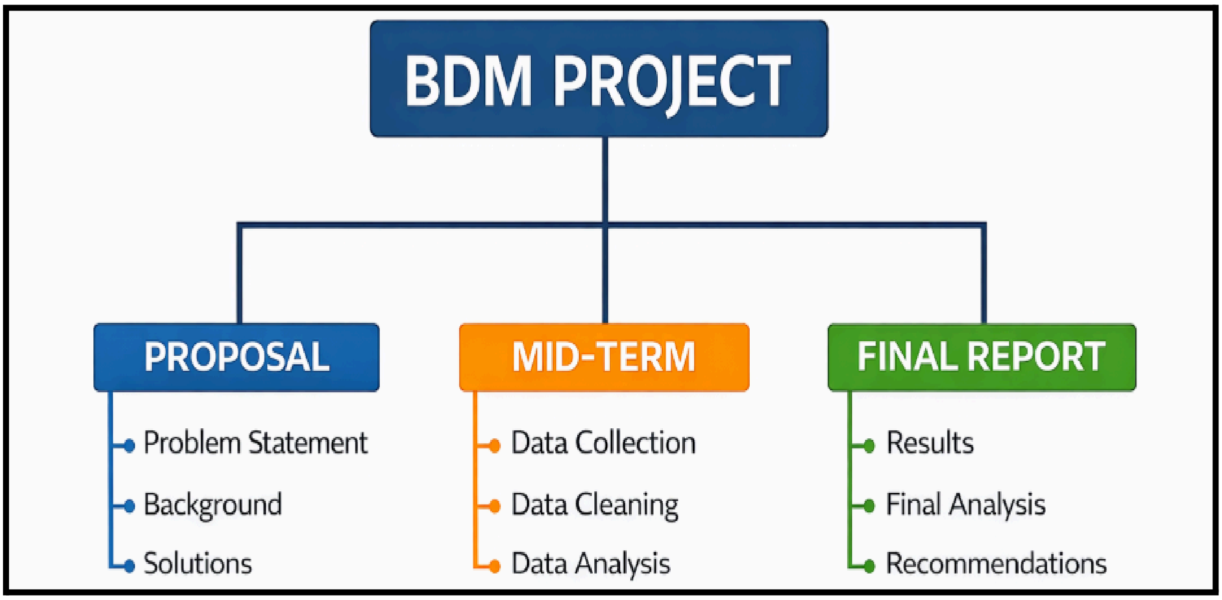


Fig .work breakdown structure –Bhavana Provision general store

7. EXPECTED OUTCOME

The expected deliverables of this project include a structured sales and customer credit dataset, analytical dashboards created using Microsoft Excel, and detailed analysis outputs generated through Python. The project will also deliver clear documentation of sales trends, customer credit behavior, and product movement patterns. A final project report summarizing insights, findings, and actionable recommendations will be submitted as part of the deliverables.

The insights generated from this project are expected to provide a clear understanding of how credit sales affect cash availability and how customer repayment behavior varies across different customer segments. Analysis of item-wise sales data will help identify fast-moving, slow-moving, and seasonal products, enabling improved inventory planning and timely replenishment decisions. These insights will support better allocation of working capital and reduction of losses due to overstocking or stockouts.

All conclusions and recommendations will be strictly data-driven, derived from systematic analysis of historical sales and customer credit records using Excel and Python. The project does not rely on assumptions or intuition but instead uses evidence-based findings to support decision-making. Overall, the expected outcome is a practical, analytics-driven framework that enhances cash flow stability, inventory efficiency, and long-term operational sustainability for the store.